AC2204 Marking Scheme

Introduction to Management Accounting · Winter 2019 · Paper 1

Examination Session and Year	Winter 2019
Module Code	AC2204
Module Title	Introduction to Management Accounting
Paper Number	Paper 1
External Examiner	Professor Ivo De Loo
Head of the Department	Professor Mark Mulcahy
Internal Examiners	Dr Margaret Healy
Duration of Paper	1.5 hours
Special Requirements	Show your workings; marks will be awarded for workings. The use of a non-programmable calculator is permitted. Separate answer books are not required.

Instructions to Candidates (as per exam paper): Answer three questions in total. Attempt Question One from Section A (compulsory, answer all parts) and TWO questions from Section B.

Own-figure rule: where a candidate makes an error at an early stage of a multi-step numerical question, full method marks are awarded for correctly applying the required technique to their own (incorrect) figure in all subsequent steps. Marks are never cascaded away for a single early error.

Marking philosophy: this is a closed-book, time-pressured examination. Academic citations are never required. Discursive answers are marked on depth of explanation and application to the specific scenario, not generic textbook recall.

Question Overview

Question	Scenario	Topics	Marks
Q1 (compulsory)	Links Ltd (Tabla & Relaxer)	CVP analysis, breakeven, margin of safety, operating leverage, multi-product WACM (new chair product, price change), CVP assumptions	40
Q2 (Section B option)	Nero Ltd	Variable (marginal) vs absorption costing, profit reconciliation	30
Q3 (Section B option)	Brighton Ltd	ABC cost hierarchy, traditional (absorption) costing vs activity-based costing	30
Q4 (Section B option)	Statement evaluation	Cost structure vs sales mix, high-low method, fixed cost per unit, conversion costs, breakeven in capital vs labour intensive companies	30

QUESTION 1

Links Ltd – 'Tabla' & 'Relaxer'

Total: 40 marks

Links Ltd manufactures kitchen furniture, with budgeted sales of the 'Tabla' kitchen table for January 2020. Candidates are required to (A) calculate breakeven, operating profit, margin of safety and operating leverage for 'Tabla' alone, (B) evaluate the introduction of a new 'Relaxer' chair product — which also changes the 'Tabla' selling price — using the WACM approach, and (C) state and explain three assumptions of CVP analysis.

Part	Requirement	Marks
(A)	Breakeven (units/revenue); operating profit, margin of safety and operating leverage at 64 units sold	14
(B)	WACM breakeven and expected profit for the Tabla + Relaxer combination	14
(C)	Three assumptions of CVP analysis	12

Question 1(A)

14 marks

Links Ltd, 'Tabla' kitchen table, January 2020: Budgeted sales price per unit €300; Budgeted variable costs per unit €125; Budgeted fixed costs per month €8,400. (i) Calculate the breakeven point, in units and in sales revenues. (ii) Calculate the operating profit, the percentage margin of safety and the degree of operating leverage, if 64 Tabla units are actually sold.

Full Model Answer

Contribution margin per unit = €300 − €125 = €175. Contribution margin ratio = €175 ÷ €300 = 58.33%.

(i) Breakeven point

Item	Value
Breakeven units (€8,400 ÷ €175)	**48 Tablas**
Breakeven revenue (48 × €300, or €8,400 ÷ 58.33%)	**€14,400**

(ii) Operating profit, margin of safety and operating leverage at 64 units sold

Item	Value
Operating profit ((64 × €175) − €8,400)	**€2,800**
Margin of safety % ((64 − 48) ÷ 64)	**25%**
Total contribution margin (64 × €175)	€11,200
Degree of operating leverage (€11,200 ÷ €2,800)	**4.0**

Mark Allocation

Marks	Criteria
12–14	Breakeven (units and revenue), operating profit, margin of safety and operating leverage all correctly calculated with clear workings.
8–11	Most figures correctly calculated with a minor error.
4–7	Some figures correctly calculated; a systematic error affects several results.
0–3	Limited understanding of CVP metrics; minimal correct workings.

Question 1(B) **14 marks**

Links Ltd is considering introducing 'Relaxer', a dining chair: variable cost €50/unit, anticipated selling price €90/unit. Management anticipates selling 4 Relaxers for each Tabla unit sold. Introducing Relaxer would increase monthly fixed costs by €1,675, and the Tabla selling price would fall by €10. (i) Determine the breakeven level of units to be sold, if Relaxer is introduced. (ii) Determine expected operating profit, if actual sales in January are 160 Relaxers and 30 Tablas.

Full Model Answer – (i) WACM breakeven

Working – revised fixed costs and contribution per unit

Item	Value
Existing fixed costs	€8,400
Additional fixed costs from introducing Relaxer	€1,675
Total fixed costs under the proposal	**€10,075**
Relaxer contribution margin per unit (€90 − €50)	€40
Revised Tabla selling price (€300 − €10)	€290
Revised Tabla contribution margin per unit (€290 − €125, variable cost unchanged)	€165

Important detail: The €10 fall in the Tabla selling price applies only to price, not to variable cost — Tabla's variable cost per unit remains €125, so its contribution margin falls from €175 to €165 (a €10 reduction, matching the price cut exactly).

Working – weighted average contribution margin (sales mix 4 Relaxers : 1 Tabla)

Item	Value
Contribution per 'basket' of 5 units (4 × €40 + 1 × €165)	€325
WACM per unit (€325 ÷ 5)	**€65.00**

Breakeven point

Item	Value
Total breakeven units (€10,075 ÷ €65)	**155 units**
Relaxer breakeven (155 × 4/5)	124 units
Tabla breakeven (155 × 1/5)	31 units

Full Model Answer – (ii) Expected operating profit at actual sales

The 4:1 sales mix ratio used in part (i) is only an assumption for the WACM breakeven calculation. Actual expected unit sales (160 Relaxers and 30 Tablas) are given directly, so profit is calculated from those actual figures.

Working – expected operating profit, January 2020

Item	€
Relaxer contribution (160 units × €40)	6,400
Tabla contribution (30 units × €165)	4,950
Total contribution margin	**11,350**
Less: Total fixed costs under the proposal	(10,075)
Expected operating profit	**1,275**

Comparison note: This expected profit of €1,275 (with the Relaxer proposal, at an actual sales mix of approximately 5.3:1 Relaxers to Tablas) can usefully be compared against the €2,800 that would have been earned from Tabla alone at 64 units (part (A)) — though a direct like-for-like comparison would require knowing how many Tablas alone would have sold without the Relaxer option, which is not given.

Mark Allocation

Marks	Criteria
13–14	WACM breakeven correctly calculated, correctly incorporating both the additional fixed costs and the reduced Tabla price/contribution; expected profit at actual sales correctly calculated from the given unit figures.
9–12	Substantially correct workings with a minor error, most commonly forgetting to reduce the Tabla contribution margin for the price cut.
5–8	WACM approach understood but applied with errors.
0–4	Limited understanding of the multi-product breakeven approach; minimal correct workings.

Question 1(C)

12 marks

State and explain three assumptions of cost-volume-profit analysis.

Full Model Answer

1. Costs can be reliably classified as fixed or variable, and behave in a linear fashion within the relevant range

Selling price, variable cost per unit, and total fixed costs are all assumed to remain constant across the range of activity being considered, so total cost and total revenue can be represented as straight lines. In practice this only holds within a limited relevant range and over a short time horizon — costs can behave differently at very different volumes (e.g. bulk discounts, or the fixed cost step-up seen in part (B), where introducing Relaxer itself requires an extra €1,675 of fixed costs).

2. The sales mix remains constant (for multi-product businesses)

Where more than one product is sold, as with Tabla and Relaxer in part (B), the WACM breakeven calculation assumes products are sold in a constant, predictable ratio (here, 4 Relaxers to 1 Tabla). In reality, the actual mix achieved may differ from the assumed mix — illustrated directly by part (B), where actual sales of 160 Relaxers to 30 Tablas is a ratio of over 5:1, not the 4:1 assumed for the breakeven calculation.

3. All units produced are sold in the same period (no inventory build-up)

CVP analysis generally assumes there is no accumulation or depletion of inventory that would otherwise distort the relationship between sales volume and profit — that is, units produced in a period are assumed to equal units sold in that period. If significant inventory levels build up or run down, the simple, single-period CVP relationship no longer directly captures the full picture (this is precisely the situation examined in the marginal versus absorption costing analysis of Question 2).

Other acceptable assumptions: Candidates may alternatively discuss: the relevant range assumption specifically (cost/revenue relationships are only valid within a defined range of activity); or that the analysis is a single-period, short-term planning tool. Any three well-explained, distinct assumptions receive full credit.

Mark Allocation

Marks	Criteria
10–12	Three clearly identified and well-explained assumptions, each accurately described and, ideally, linked to the specific scenarios elsewhere in this question.
7–9	Three assumptions identified with reasonable explanation.
4–6	Two assumptions identified with reasonable explanation, or three assumptions stated with minimal explanation.
0–3	Minimal or no relevant assumptions identified.

QUESTION 1 TOTAL — 40 MARKS

QUESTION 2

Nero Ltd

Total: 30 marks

Nero Ltd commenced operations on 1 November 2018, manufacturing and selling a single product. Candidates are required to (A) calculate unit cost and closing stock, (B) calculate operating profit under both variable and absorption costing, and (C) reconcile the resulting profit figures.

Part	Requirement	Marks
(A)	Cost per unit and value of closing stock – variable and absorption costing	6
(B)	Operating profit – variable and absorption costing	18
(C)	Profit reconciliation, with explanation	6

Question 2(A) **6 marks**

Nero Ltd, year ending 31 October 2019: Sales 4,500 units; Production 5,100 units; Selling price €40.00; Variable manufacturing costs (total) €114,750; Fixed manufacturing costs (total) €19,890; Variable marketing costs (total) €19,125; Fixed marketing costs (total) €20,000. (A) Calculate the cost per unit and the value of closing stock using (i) Variable costing and (ii) Absorption costing.

Full Model Answer

This is Nero Ltd's first year, so there is no opening stock. Closing stock = Production − Sales = 5,100 − 4,500 = 600 units. Variable manufacturing cost per unit = €114,750 ÷ 5,100 = €22.50. Fixed manufacturing overhead per unit (absorption costing only) = €19,890 ÷ 5,100 = €3.90.

Cost per unit and closing stock value

Item	Variable costing	Absorption costing
Unit product cost	€22.50	€26.40 (€22.50 + €3.90)
Closing stock value (600 units)	€13,500	€15,840

Mark Allocation

Marks	Criteria
5–6	Both unit costs and both closing stock values correctly calculated.
3–4	Substantially correct with a minor computational error.
1–2	Some correct workings but a fundamental error in one method.
0	No meaningful correct workings.

Question 2(B) **18 marks**

Calculate the operating profit using (i) Variable costing and (ii) Absorption costing.

Full Model Answer – (i) Variable costing

Income statement – variable costing

Item	€
Sales revenue (4,500 units × €40.00)	180,000
Less: Variable cost of goods sold (4,500 units × €22.50)	(101,250)
Less: Variable marketing costs	(19,125)
Contribution margin	**59,625**
Less: Fixed manufacturing costs	(19,890)
Less: Fixed marketing costs	(20,000)
Operating profit (variable costing)	**19,735**

Full Model Answer – (ii) Absorption costing

Income statement – absorption costing

Item	€
Sales revenue (4,500 units × €40.00)	180,000
Less: Cost of goods sold (4,500 units × €26.40)	(118,800)
Gross profit	**61,200**
Less: Variable marketing costs	(19,125)
Less: Fixed marketing costs	(20,000)
Operating profit (absorption costing)	**22,075**

Mark Allocation

Marks	Criteria
15–18	Both income statements correctly prepared, with contribution margin and gross profit subtotals clearly shown and correct final operating profit figures.
11–14	Substantially correct with a minor computational error in one figure.
6–10	One method correct; the other contains significant errors.
0–5	Limited understanding of either costing method; minimal correct workings.

Question 2(C)

6 marks

Reconcile the profit figures from Part (B) above. In your answer, briefly explain why this difference arises.

Full Model Answer

Why the difference arises

Absorption costing defers fixed manufacturing overhead into closing stock (as part of the unit product cost), while variable costing expenses fixed manufacturing overhead in full as a period cost, regardless of how many units remain unsold. Because production (5,100 units) exceeded sales (4,500 units) this year, more fixed overhead was deferred into closing stock under absorption costing than was expensed under variable costing, resulting in a higher reported profit under absorption costing.

Profit reconciliation

Item	€
Operating profit – variable costing	19,735
Add: Fixed manufacturing overhead carried forward in closing stock (600 units × €3.90)	2,340
Less: Fixed manufacturing overhead carried forward in opening stock (none – first year of trading)	0
Operating profit – absorption costing	**22,075**

Mark Allocation

Marks	Criteria
5–6	Clear explanation of the fixed overhead deferral in stock under absorption costing, why production exceeding sales causes profit to differ, and a correct, clearly presented profit reconciliation.
3–4	Good explanation with a correct or substantially correct reconciliation; some lack of clarity or a minor computational error.
1–2	Some understanding shown of why the figures differ, but explanation is generic or the reconciliation is materially incomplete.
0	No meaningful explanation or reconciliation provided.

QUESTION 2 TOTAL — 30 MARKS

QUESTION 3

Brighton Ltd

Total: 30 marks

Brighton Ltd makes two types of storage units (Basic and Deluxe models) and currently absorbs production overheads using a machine hour rate, while considering activity-based costing. Candidates are required to (A) list and explain the levels of the ABC cost hierarchy with examples, and (B) recalculate product costs under both traditional and activity-based costing.

Part	Requirement	Marks
(A)	List and explain the levels of the ABC hierarchy, with an example of each	8
(B)	Total costs and cost per unit – traditional costing and activity-based costing	22

Question 3(A) **8 marks**

List and explain the levels of activities in an Activity-based costing (ABC) hierarchy, giving an example of each.

Full Model Answer

The ABC cost hierarchy

Level	Description	Example
Unit-level	Costs incurred once for every single unit of output; vary directly and proportionally with the number of units produced.	Direct materials and direct labour consumed by each individual storage unit produced.
Batch-level	Costs incurred once for a batch or production run, regardless of how many units are in the batch.	Machine set-up costs incurred each time a new production run begins — as in this question, where Basic units are produced in batches of 50 and Deluxe units in batches of 25.
Product-sustaining level	Costs incurred to support a particular product line as a whole, regardless of the number of units or batches produced.	Costs of maintaining a product-specific design, engineering specification, or dedicated tooling for the Deluxe model.
Facility-sustaining level	Costs incurred to support the organisation as a whole; cannot be traced to any specific unit, batch, or product line.	Rent of the factory building, general factory management salaries, and plant security — incurred regardless of which products are made or in what volume.

Mark Allocation

Marks	Criteria
7–8	All four levels of the ABC hierarchy correctly identified and described; a specific, plausible example given for each level.
5–6	Most levels correctly identified and described; examples given but one or two less specific.
2–4	Some levels of the hierarchy identified; descriptions or examples generic or incomplete.
0–1	Minimal or no understanding of the ABC cost hierarchy.

Question 3(B) **22 marks**

Brighton Ltd, Basic Model: 2,500 units, €24.50 DM/unit, €28.30 DL/unit, 50 units/batch, 4 accessories/unit, 5 MH/unit. Deluxe Model: 1,000 units, €53.00 DM/unit, €31.80 DL/unit, 25 units/batch, 17 accessories/unit, 15 MH/unit. Cost pools: Machining costs €206,250 (machine hours); Set-up costs €9,000 (production runs); Accessories €43,740 (number of accessories). (B) Calculate the total costs and the cost per unit for each product using (i) Traditional (absorption) costing and (ii) Activity-based costing.

Full Model Answer – (i) Traditional (absorption) costing

Total production overheads = €206,250 + €9,000 + €43,740 = €258,990. Total machine hours = (2,500 × 5) + (1,000 × 15) = 12,500 + 15,000 = 27,500. Overhead absorption rate = €258,990 ÷ 27,500 = €9.42 per machine hour (rounded).

Traditional (absorption) costing – cost per unit

Item	Basic Model	Deluxe Model
Direct materials	24.50	53.00
Direct labour	28.30	31.80
Overhead (MH/unit × €9.42)	47.09	141.27
Total cost per unit	**99.89**	**226.07**

Traditional (absorption) costing – total costs

Item	Basic Model (€)	Deluxe Model (€)
Units produced	2,500	1,000
Total cost (unit cost × units)	**249,725**	**226,070**

Rounding note: The overhead rate of €9.42 per machine hour does not divide evenly; figures throughout are rounded to the nearest cent, so totals may differ very slightly (a few euro) from €258,990 due to rounding. Candidates working with more decimal places, or rounding at a different stage, receive full own-figure credit provided the method is correct.

Full Model Answer – (ii) Activity-based costing

Working – number of production runs (batches) and total accessories

Item	Basic Model	Deluxe Model	Total
Production runs (units ÷ batch size)	2,500 ÷ 50 = 50	1,000 ÷ 25 = 40	90
Total accessories (units × accessories/unit)	2,500 × 4 = 10,000	1,000 × 17 = 17,000	27,000

Working – cost pools and activity rates

Cost pool	Pool amount (€)	Cost driver	Total activity	Rate
Machining costs	206,250	Machine hours	27,500 MH	€7.50 per MH
Set-up costs	9,000	Production runs	90 runs	€100.00 per run
Accessories	43,740	No. of accessories	27,000 accessories	€1.62 per accessory
Total	**258,990**			

Working – overhead allocated to each product

Cost pool	Basic Model (€)	Deluxe Model (€)
Machining (total MH × €7.50)	93,750	112,500
Set-up (runs × €100)	5,000	4,000
Accessories (total accessories × €1.62)	16,200	27,540
Total overhead	**114,950**	**144,040**

Check: Total overhead allocated = €114,950 + €144,040 = €258,990, agreeing with the total production overhead given.

Activity-based costing – total costs and cost per unit

Item	Basic Model	Deluxe Model
Direct materials (total)	61,250	53,000
Direct labour (total)	70,750	31,800
Overhead (from above)	114,950	144,040
Total cost	**246,950**	**228,840**
Units produced	2,500	1,000
Cost per unit	**98.78**	**228.84**

Comparison: ABC slightly reduces Basic Model's unit cost (€99.89 → €98.78) and slightly increases Deluxe Model's (€226.07 → €228.84). Deluxe, despite lower production volume, uses substantially more accessories per unit (17 vs 4) and requires proportionately more frequent set-ups per unit produced (40 runs for only 1,000 units, i.e. one run per 25 units, versus one run per 50 units for Basic), both of which the traditional machine-hour-only allocation understates.

Mark Allocation

Marks	Criteria
18–22	Traditional costing correctly calculated; production runs and total accessories correctly derived; all three ABC cost pool rates correctly calculated; overhead correctly allocated; ABC total cost and cost per unit correct for both products.
13–17	Substantially correct with minor computational errors in one or two rates or allocations.
7–12	Traditional costing correct; ABC approach understood but the batch/accessory derivation or rates/allocations contain multiple errors.
0–6	Limited understanding of either costing method; minimal correct workings.

QUESTION 3 TOTAL — 30 MARKS

QUESTION 4

Statement Evaluation

Total: 30 marks

Five independent statements covering cost structure vs sales mix, the high-low method, fixed cost per unit behaviour, conversion cost calculations, and breakeven point in capital vs labour intensive companies. For each statement, candidates must state whether they agree or disagree, with supporting calculations and explanation.

Part	Topic	Marks
(A)	Cost structure vs sales mix	6
(B)	High-low method cost estimate	6
(C)	Fixed cost per unit as activity increases	6
(D)	Azure Ltd – manufacturing overhead from conversion cost percentage	6
(E)	Breakeven point: capital vs labour intensive	6

Question 4(A)

6 marks

Cost structure refers to the relative proportions in which an organisation's products are sold.

Full Model Answer

DISAGREE

The description given in the statement is the definition of **sales mix**, not cost structure.

> ### What cost structure actually is
> Cost structure refers to the relative proportion of fixed costs and variable costs within an organisation's total costs. A cost structure weighted towards fixed costs (capital intensive) produces different operating leverage, margin of safety, and breakeven characteristics than one weighted towards variable costs (labour intensive), as examined further in part (E) of this question.

> ### What the statement is actually describing
> The relative proportions in which an organisation's different products are sold is the definition of **sales mix**, which is central to multi-product breakeven analysis using the weighted average contribution margin approach, as applied to Links Ltd's Tabla and Relaxer products in Question 1.

Mark Allocation

Marks	Criteria
5–6	Correct disagreement with an accurate definition of cost structure and a correct identification that the statement's description actually matches sales mix.
3–4	Correct disagreement with a reasonable definition of cost structure; the sales mix point not explicitly identified.
1–2	Correct disagreement but with an inaccurate or vague definition of cost structure.
0	Incorrect verdict (agrees with the statement) or no meaningful explanation.

Question 4(B) **6 marks**

Monthly activity levels and associated costs recorded in 2018: HIGH: April, 2,100 units, total cost €62,390. LOW: August, 900 units, total cost €35,030. If it is planned to produce 1,900 units in December 2019, then based on the cost function for 2018, the expected cost is €57,830.

Full Model Answer

AGREE

The expected cost of €57,830 given in the statement is correct.

Working – high-low method

Item	Value
Variable cost per unit ((€62,390 − €35,030) ÷ (2,100 − 900))	(€27,360 ÷ 1,200) = €22.80 per unit
Fixed cost (at high point: €62,390 − (2,100 × €22.80))	€14,510
Fixed cost (at low point: €35,030 − (900 × €22.80)) – check	€14,510
Cost function	Total cost = €14,510 + (€22.80 × units)

Estimated cost at 1,900 units (December 2019)

Item	Value
€14,510 + (€22.80 × 1,900)	€14,510 + €43,320
Estimated cost	**€57,830**

Mark Allocation

Marks	Criteria
5–6	Correct agreement; variable cost per unit and fixed cost correctly calculated and cross-checked at both points; correct cost function applied to confirm €57,830.
3–4	Correct agreement with a minor computational error, or the cross-check at the second point omitted.
1–2	High-low method attempted with the correct general approach but a significant computational error.
0	Incorrect verdict (disagrees with the statement) or no meaningful workings.

Question 4(C)

6 marks

Fixed costs per unit remain constant as the level of activity increases within the relevant range.

Full Model Answer

DISAGREE

It is **total** fixed costs that remain constant within the relevant range, not fixed cost **per unit**. Fixed cost per unit actually decreases as activity increases.

> **Why fixed cost per unit falls, not stays constant**
> Fixed cost per unit is calculated as total fixed cost divided by the number of units of activity. Since total fixed cost is unchanged within the relevant range, spreading that same total over more units as volume rises causes the fixed cost per unit to fall. For example, from Question 1, Links Ltd's €8,400 monthly fixed costs give a fixed cost per Tabla of €175 at 48 units (breakeven), but only €131.25 per unit at 64 units sold — the total is unchanged, but the per-unit figure falls as volume rises.

Mark Allocation

Marks	Criteria
5–6	Correct disagreement with clear explanation of why fixed cost per unit falls (not remains constant) as volume rises, correctly distinguishing total fixed cost (which does remain constant) from fixed cost per unit, with an illustrative example.
3–4	Correct disagreement with reasonable explanation; illustrative example missing.
1–2	Correct disagreement but with minimal or unclear explanation.
0	Incorrect verdict (agrees with the statement) or no meaningful explanation.

Question 4(D) 6 marks

At Azure Ltd, manufacturing overhead is 40% of its total conversion costs, direct labour is €18,000 and direct materials are €24,000. On this basis, the manufacturing overhead must be €30,000.

Full Model Answer

DISAGREE

The correct manufacturing overhead figure is €12,000, not the €30,000 stated. This is a genuine computational error in the statement, not merely a different (also valid) way of reading the question.

> **Key point – direct materials is not used in this calculation**
> Conversion cost is defined as direct labour plus manufacturing overhead; it deliberately **excludes** direct materials. The €24,000 direct materials figure given in the statement is not required for this calculation at all — including it (for example, by treating €42,000 of direct labour and materials combined as the base) is a common error that leads to the incorrect €30,000 figure.

Working – correct calculation

Item	Value
Let manufacturing overhead (MOH) = 40% of conversion cost	MOH = 0.40 × (DL + MOH)
MOH = 0.40 × €18,000 + 0.40 × MOH	MOH − 0.40 MOH = €7,200
0.60 × MOH = €7,200	MOH = €7,200 ÷ 0.60
Correct manufacturing overhead	**€12,000**

Check

Item	€
Direct labour	18,000
Manufacturing overhead	12,000
Conversion cost	**30,000**
Manufacturing overhead as % of conversion cost (€12,000 ÷ €30,000)	40% – confirms the correct figure

Where the incorrect figure likely came from: If MOH were €30,000, conversion cost would be €18,000 + €30,000 = €48,000, and MOH would represent 30,000 ÷ 48,000 = 62.5% of conversion cost, not 40% — confirming €30,000 does not satisfy the condition given and is simply incorrect.

Mark Allocation

Marks	Criteria
5–6	Correct disagreement; conversion cost correctly defined as excluding direct materials; algebraic working correctly solved to give €12,000, with a check confirming the 40% relationship and disproving the €30,000 claim.
3–4	Correct disagreement and correct final figure of €12,000, but the check disproving €30,000 not shown.
1–2	Correct verdict stated but workings contain an error, or direct materials incorrectly included in the calculation.
0	Incorrect verdict (agrees with the statement) or no meaningful workings.

Question 4(E)

6 marks

The breakeven point in a capital intensive company will tend to be lower than in a labour intensive company with the same total sales revenue and total expenses.

Full Model Answer

DISAGREE

The breakeven point (in revenue terms) will tend to be **higher**, not lower, in a capital intensive company compared to a labour intensive company with the same total sales revenue and total expenses.

Why capital intensity raises the breakeven point

A capital-intensive company has a cost structure weighted more heavily towards fixed costs (e.g. depreciation on machinery) and correspondingly lower variable costs, compared to a labour-intensive company with the same total revenue and total costs. Breakeven revenue = Fixed costs ÷ Contribution margin ratio. Although the capital-intensive company's contribution margin ratio is higher (since its variable cost per euro of sales is lower), its fixed cost base is also proportionately much larger, and the fixed cost effect dominates: a larger absolute fixed cost requires a larger absolute contribution to cover it, giving the capital-intensive company a **higher** breakeven point.

Illustration (for the same €100 revenue and €80 total costs in both companies)

Item	Capital intensive	Labour intensive
Fixed costs	€60	€20
Variable costs	€20	€60
Contribution margin ratio	80%	40%
Breakeven revenue (Fixed ÷ CM ratio)	**€75**	**€50**

Consistency with earlier questions: This is consistent with the established relationship that capital-intensive companies have higher operating leverage and a lower margin of safety than labour-intensive companies with the same revenue and costs — a higher breakeven point (relative to sales) is precisely what produces that lower margin of safety.

Mark Allocation

Marks	Criteria
5–6	Correct disagreement with a clear explanation of why a higher fixed cost proportion raises the breakeven point, ideally supported by a numerical illustration, and consistency with the operating leverage/margin of safety relationship.
3–4	Correct disagreement with reasonable explanation but no numerical illustration.
1–2	Correct verdict stated but with minimal or unclear explanation.
0	Incorrect verdict (agrees with the statement) or no meaningful explanation.

QUESTION 4 TOTAL — 30 MARKS